

Air University



Computer Networks (Lab)

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Lab Task # 01

1. Is the frame an outgoing or an incoming frame?

The frame is an **outgoing frame**. The selected frame (marked with a blue arrow and highlighted) has a **Source IP address** of **192.168.1.124** and a **Destination IP address** of **54.231.129.165**.

No.	Time	Source	Destination	Protocol	Length Info
2900	59.711465	192.168.1.124	54.231.129.165	HTTP	330 GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
3202	60.023327	54.231.129.165	192.168.1.124	HTTP	878 HTTP/1.1 200 OK (application/pkix-cert)
37821	1592.338381	192.168.1.124	92.123.159.240	HTTP	288 GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBTjrjryd
37825	1592.442605	92.123.159.240	192.168.1.124	OCSP	622 Response
38768	1649.707610	192.168.1.124	199.232.82.172	HTTP	336 GET /msdownload/update/v3/static/trustedr
38770	1649.895937	199.232.82.172	192.168.1.124	HTTP	257 HTTP/1.1 304 Not Modified

2. What is the source IP address of the network-layer header in the frame?

The source IP address is **192.168.1.124**. This is found in the **Source** column of the selected packet. The **network-layer** (Layer 3) header in this case is the **Internet Protocol (IP)** header, and the IP addresses are what you see in the Source and Destination columns.

No.	Time	Source	Destination	Protocol	Length Info
2900	59.711465	192.168.1.124	54.231.129.165	HTTP	330 GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
3202	60.023327	54.231.129.165	192.168.1.124	HTTP	878 HTTP/1.1 200 OK (application/pkix-cert)
37821	1592.338381	192.168.1.124	92.123.159.240	HTTP	288 GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBTjrjrydR
37825	1592.442605	92.123.159.240	192.168.1.124	OCSP	622 Response
38768	1649.707610	192.168.1.124	199.232.82.172	HTTP	336 GET /msdownload/update/v3/static/trustedr/
38770	1649.895937	199.232.82.172	192.168.1.124	HTTP	257 HTTP/1.1 304 Not Modified

3. What is the destination IP address of the network-layer header in the frame?

The destination IP address is **54.231.129.165**. This is found in the **Destination** column of the selected packet. The **network-layer** (Layer 3) header is the **Internet Protocol (IP)** header.

No.	Time	Source	Destination	Protocol	Length Info
2900	59.711465	192.168.1.124	54.231.129.165	HTTP	330 GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
3202	60.023327	54.231.129.165	192.168.1.124	HTTP	878 HTTP/1.1 200 OK (application/pkix-cert)
37821	1592.338381	192.168.1.124	92.123.159.240	HTTP	288 GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBTjrjryc
37825	1592.442605	92.123.159.240	192.168.1.124	OCSP	622 Response
38768	1649.707610	192.168.1.124	199.232.82.172	HTTP	336 GET /msdownload/update/v3/static/trustedr
38770	1649.895937	199.232.82.172	192.168.1.124	HTTP	257 HTTP/1.1 304 Not Modified

4. What is the total number of bytes in the whole frame?

The total number of bytes in the whole frame (as it appeared on the wire) is **330 bytes**. This information is displayed right next to the **Frame** layer details in the middle pane.

```

> Frame 2900: 330 bytes on wire (2640 bits), 330 bytes captured (2640 bits) on interface \Device\NPF_{B9750D9C-BB54-40EF-AF4A-5C50EB21BB96}, id 0
> Ethernet II, Src: Intel_1f:81:eb (24:41:8c:1f:81:eb), Dst: TendaTechnol_5d:2c:30 (08:40:f3:5d:2c:30)
> Internet Protocol Version 4, Src: 192.168.1.124, Dst: 54.231.129.165
> Transmission Control Protocol, Src Port: 60878, Dst Port: 80, Seq: 1, Ack: 1, Len: 276
> Hypertext Transfer Protocol

```

5. What is the number of bytes in the Ethernet (data-link layer) header?

The information visible under the **Ethernet II** layer directly reveals the header's components:

1. **Destination MAC Address:** TendaTechnol_5d:2c:30 is **6 bytes**.
2. **Source MAC Address:** Intel_1f:81:eb is **6 bytes**.
3. **Type:** IPv4 (0x0800) is **2 bytes**.

Calculation:

Ethernet II Header Size = Destination MAC + Source MAC + Type Field

Ethernet II Header Size = 6 bytes + 6 bytes + 2 bytes = 14 bytes.

The total size of the Ethernet (data-link layer) header is **14 bytes**.

```

▼ Ethernet II, Src: Intel_1f:81:eb (24:41:8c:1f:81:eb), Dst: TendaTechnol_5d:2c:30 (08:40:f3:5d:2c:30)
  ► Destination: TendaTechnol_5d:2c:30 (08:40:f3:5d:2c:30)
  ► Source: Intel_1f:81:eb (24:41:8c:1f:81:eb)
    Type: IPv4 (0x0800)
    [Stream index: 1]

```

6. What is the number of bytes in the IP header?

The number of bytes in the IP header is **20 bytes**. This is explicitly stated in the "**Header Length**" field:

.... 0101 = Header Length: 20 bytes (5)

```

Internet Protocol Version 4, Src: 192.168.1.124, Dst: 54.231.129.165
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ► Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 316
    Identification: 0x0906 (2310)
  ► 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.124
    Destination Address: 54.231.129.165
    [Stream index: 29]

```

7. What is the number of bytes in the TCP header?

The number of bytes in the TCP header is **20 bytes**. This is explicitly stated in the "**Header Length**" field shown within the TCP details: 0101 = Header Length: 20 bytes (5)

```

▼ Transmission Control Protocol, Src Port: 60878, Dst Port: 80, Seq: 1, Ack: 1, Len: 276
  Source Port: 60878
  Destination Port: 80
  [Stream index: 53]
  [Stream Packet Number: 4]
  ► [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 276]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 2020369044
  [Next Sequence Number: 277 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3017928776
  0101 .... = Header Length: 20 bytes (5)
  ► Flags: 0x018 (PSH, ACK)
  Window: 257
  [Calculated window size: 65792]
  [Window size scaling factor: 256]
  Checksum: 0x7bdf [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ► [Timestamps]
  ► [SEQ/ACK analysis]
  TCP payload (276 bytes)

```

8. What is the total bytes in the message (at the application layer)?

The application layer message size is the **Total Frame Length** minus all the lower-layer header lengths. The simplest way to find this is to use the **TCP Segment Length (Len)** or the **TCP Payload** length, as the application data is the payload of the TCP segment. The total bytes in the message (at the application layer) is **276 bytes**.

```
Transmission Control Protocol, Src Port: 60878, Dst Port: 80, Seq: 1, Ack: 1, Len: 276
  Source Port: 60878
  Destination Port: 80
  [Stream index: 53]
  [Stream Packet Number: 4]
  ▶ [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 276]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 2020369044
  [Next Sequence Number: 277 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3017928776
  0101 .... = Header Length: 20 bytes (5)
  ▶ Flags: 0x018 (PSH, ACK)
  Window: 257
  [Calculated window size: 65792]
  [Window size scaling factor: 256]
  Checksum: 0x7bdf [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ [Timestamps]
  ▶ [SEQ/ACK analysis]
  TCP payload (276 bytes)
```

THE END
